OPTIMISATION

1. a) A direct computation yields

$$\nabla f(x_1,x_2) = 2 \begin{bmatrix} x_1(1-\sin x_1^2) \\ x_2 \end{bmatrix}$$

and

$$\nabla^2 f(x_1,x_2) = 2 \begin{bmatrix} \star(x_1) & 0 \\ 0 & 1 \end{bmatrix},$$

where $\star(x_1) = (1-\sin x_1^2) - 2x_1^2 \cos x_1^2$. Evaluating $\star(x_1)$ at $\pm\sqrt{2k\pi}$, for k nonzero integer, yields $\star(\pm\sqrt{2k\pi}) = 1-2|2k\pi|$, which is negative for all nonzero k, hence $\nabla^2 f$ is not positive definite for all (x_1,x_2), hence the function is not convex.

From the expression of ∇f one has that $(0,0)$ is a stationary point such that $\nabla^2 f(0,0) > 0$, hence $(0,0)$ is a local strict minimizer. In addition, all points $(\bar{x}_1,\bar{x}_2)$ such that $(1-\sin\bar{x}_1^2) = 0$ and $\bar{x}_2 = 0$ are stationary points, that is the function f has the stationary points $(\pm\sqrt{k\pi},0)$, for k integer, that is infinitely many stationary points. [4 marks]

b) Note that

$$\nabla f(x_1,x_2) = \begin{bmatrix} 4x_1 - x_2 - 3 + 2e^{2x_1+x_2} \\ -x_1 + 2x_2 + e^{2x_1+x_2} \end{bmatrix},$$

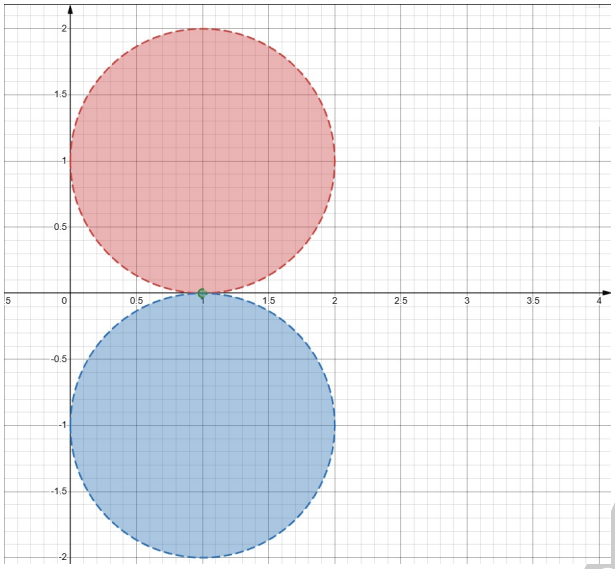
hence $\nabla f(0,0) = [-1,\ 1]^\top$. A descent direction for f at x_0 is $d = -\nabla f(0,0)$. Note now that

$$f(\overbrace{x_0+\alpha d}^{x_N}) - f(x_0) = 2\alpha^2 + \alpha + \alpha^2 - 3\alpha + e^\alpha - 1 \approx -\alpha + \frac{7}{2}\alpha^2 + \cdots,$$

hence one could select, for example, $\alpha = 0.1$ to have $f(x_N) < f(x_0)$. Note that any $\alpha \in (0, 0.281655...)$ gives the indicated inequality. [4 marks]

c) i) The admissible set is the intersection of two disks (red and blue, with the boundary – dashed – included), as indicated in the graph below. In particular the admissible set is only the point $(1,0)$ (the "green" dot in the graph).

Answers



The gradients of the active constraints (both are active at $(1,0)$) are

$$\nabla h_1(1,0) = \begin{bmatrix} 0 \\ -2 \end{bmatrix}, \qquad \nabla h_2(1,0) = \begin{bmatrix} 0 \\ 2 \end{bmatrix},$$

which are linearly dependent, hence the point $(1,0)$ is not a regular point. [4 marks]

ii) The optimal solution of the problem is the point $(1,0)$ – since the admissible set contains only one point. The Lagrangian of the problem is

$$\begin{aligned} L(x_1,x_2,\rho_1,\rho_2) &= x_1 + \rho_1((x_1-1)^2+(x_2-1)^2-1)+ \\ &\quad \rho_2((x_1-1)^2+(x_2+1)^2-1), \end{aligned}$$

and the first two necessary conditions of optimality are

$$0 = 1+2\rho_1(x_1-1)+2\rho_2(x_1-1), \qquad 0 = 2\rho_1(x_2-1)+2\rho_2(x_2+1).$$

Replacing $(1,0)$ in the above equations yields

$$0 = 1, \qquad 0 = -2\rho_1+2\rho_2,$$

which do not have any solution, hence the optimal point does not satisfy the necessary conditions of optimality. [4 marks]

d) Eliminating the equality constraint by replacing x_2 with $x_1 - 1$ yields

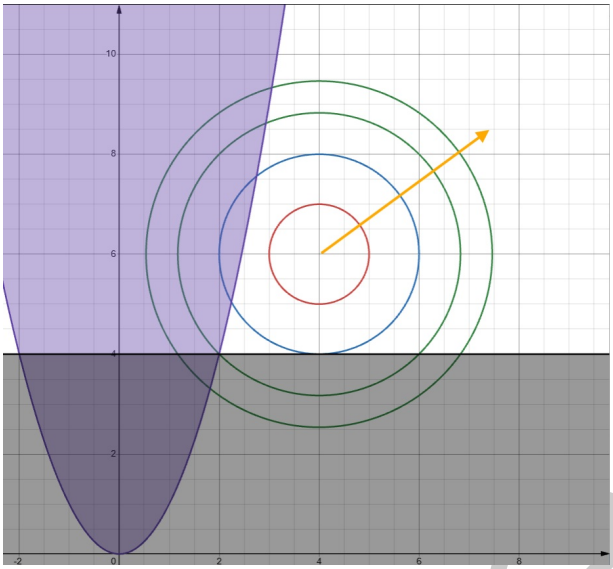
$$\min_{x_1} x_1^2 - (x_1-1)^2,$$
$$x_1 \geq 1.$$

Note that $x_1^2 - (x_1-1)^2 = 2x_1 - 1$, which is minimized (in the admissible set) by $x_1 = 1$. Note also that since the *reduced* objective function is linear and monotonically increasing, $x_1 = 1$ is a (global) strict minimizer.

The same result is obtained by eliminating x_1 using the equation $x_1 = x_2 + 1$, which yields the constraint $x_2 \geq 0$. [4 marks]

Answers

2. a) The level lines of the cost function (the circles centered at $(4,6)$) and the admissible set (the intersection of the shaded areas) are sketched in the graph below. The function f increases along the direction of the orange arrow.



[4 marks]

b) All interior points are regular points. Note now that

$$\nabla h_1 = \begin{bmatrix} 2x_1 \\ -1 \end{bmatrix}, \qquad \nabla h_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

These are linearly dependent only at $x_1 = 0$, where however only one constraint may be active. Hence, all points are regular points.

The optimal solution is at the top-right corner of the admissible set, that is the point $(2,4)$. At this point one has $f(2,4) = 8$. [4 marks]

c) The Lagrangian of the problem is

$$L(x_1,x_2,\rho_1,\rho_2) = (x_1-4)^2 + (x_2-6)^2 + \rho_1(x_1^2 - x_2) + \rho_2(x_2-4).$$

The first order necessary conditions of optimality are

$$0 = 2(x_1-4) + 2\rho_1 x_1, \qquad 0 = 2(x_2-6) + \rho_2$$

$$x_1^2 - x_2 \le 0, \qquad x_2 - 4 \le 0, \qquad \rho_1 \ge 0, \qquad \rho_2 \ge 0,$$

$$\rho_1(x_1^2 - x_2) = 0, \qquad \rho_2(x_2-4) = 0.$$

Using the complementarity conditions yields four cases.

i) $\rho_1 = 0$ and $\rho_2 = 0$. This yields $(x_1,x_2) = (4,6)$ which is not feasible.

ii) $\rho_1 > 0$ and $\rho_2 = 0$. This yields $x_2 = 6$ which is not feasible.

iii) $\rho_1 = 0$ and $\rho_2 > 0$. This yields $x_1 = 4$ and $x_2 = 4$, which is not feasible.

iv) $\rho_1 > 0$ and $\rho_2 > 0$. This yields $x_2 = 4$, $x_1 = 2$, $\rho_1 = 1$, and $\rho_2 = 4$. Note that $x_2 = 4$, $x_1 = -2$ yields a negative ρ_1, hence it does not satisfy the necessary conditions.

[8 marks]

Answers

d) To check the indicates sufficient condition, note that its left had side is given by

$$\begin{bmatrix} -2 & -4 \end{bmatrix} \begin{bmatrix} x_1 - 2 \\ x_2 - 4 \end{bmatrix}.$$

Note now that $x_1 - 2$ and $x_2 - 4$ are non-positive for all points in the admissible set, hence (in the admissible set)

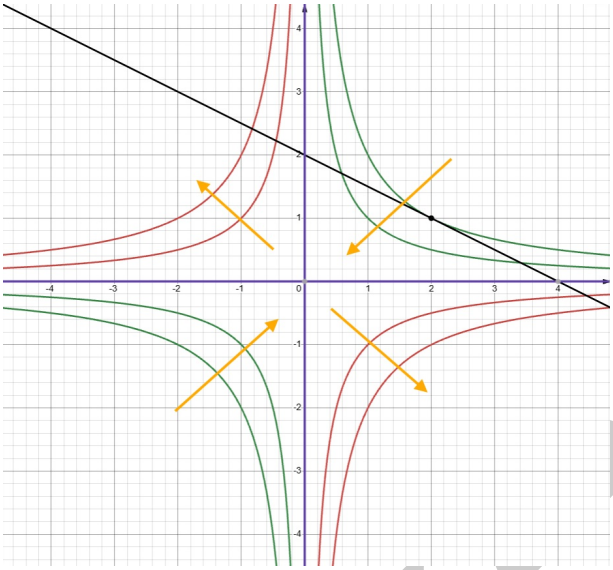
$$-2 \overbrace{(x_1 - 2)}^{\leq 0} - 4 \overbrace{(x_1 - 4)}^{\leq 0} \geq 0,$$

which shows that the candidate optimal point is indeed a (global) minimizer.

[4 marks]

Answers

3. a) The level lines of the cost function and the admissible set are sketched in the graph below. The admissible set is the black line; the level lines for $f = 0$ are in violet; the level lines for $f < 0$ are in green; the level lines for $f > 0$ are in red. The function f increases along the direction of the orange arrows. The optimal solution is the point $P_\star = (2,1)$, indicated with a small black dot.



[4 marks]

b) The Lagrangian for the problem is

$$L(x_1, x_2, \lambda) = -x_1 x_2 + \lambda(x_1 + 2x_2 - 4).$$

The necessary conditions of optimality are

$$0 = -x_2 + \lambda, \qquad 0 = -x_1 + 2\lambda, \qquad x_1 + 2x_2 - 4 = 0.$$

These yield the only candidate point $(x_1, x_2, \lambda) = (2, 1, 1)$.

The second order sufficient conditions give (note that the equality constraint is active)

$$s^\top \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} s$$

with $s^\top = \alpha \begin{bmatrix} 2 & -1 \end{bmatrix}$ and $\alpha \neq 0$, yielding

$$s^\top \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} s = 4\alpha^2,$$

hence $P_\star$ is a local strict minimizer. [4 marks]

c) The sequential penalty function is

$$F_\varepsilon(x_1, x_2) = -x_1 x_2 + \frac{1}{\varepsilon}(x_1 + 2x_2 - 4)^2,$$

and its stationary points are the solution of the equations

$$0 = -x_2 + \frac{2}{\varepsilon}(x_1 + 2x_2 - 4), \qquad 0 = -x_1 + \frac{4}{\varepsilon}(x_1 + 2x_2 - 4).$$

These yield the unique stationary point

$$P_\varepsilon = \left(2\,\frac{8}{8-\varepsilon}, \frac{8}{8-\varepsilon}\right),$$

which, as $\varepsilon \to 0$, converges to $P_\star$. [4 marks]

Answers

d) The exact penalty function is

$$G(x) = -x_1x_2 + \lambda(x)(x_1 + 2x_2 - 4) + \frac{1}{\varepsilon}(x_1 + 2x_2 - 4)^2,$$

with

$$\lambda(x) = \frac{2x_1 + x_2}{5}.$$

Note now that

$$\nabla G(x) = \begin{bmatrix} -x_2 + \frac{2}{5}(x_1 + 2x_2 - 4) + \frac{1}{5}(2x_1 + x_2) + \frac{2}{\varepsilon}\cdots \\ -x_1 + \frac{1}{5}(x_1 + 2x_2 - 4) + \frac{2}{5}(2x_1 + x_2) + \frac{2}{\varepsilon}\cdots \end{bmatrix},$$

where the "ε" terms are not important, and that the equation $\nabla G = 0$ has the unique solution $x_1 = 2$, $x_2 = 1$, which is the optimal solution of the considered problem. [8 marks]

Answers

4. a) Newton's iteration for $x \leq -1$ is

$$x_{k+1} = x_k - \frac{2\,(x_k - 1)}{2} = 1,$$

whereas Newton's iteration for $x \geq 1$ is

$$x_{k+1} = x_k - \frac{2\,(x_k + 1)}{2} = -1.$$

Hence, one has $x_0 = 1$, $x_1 = -1$, $x_2 = 1$ and so on. [4 marks]

b) i) Note that

$$\nabla f(a,b,c) = \begin{bmatrix} 2ab + c^2 \\ 2bc + a^2 \\ 2ca + b^2 \end{bmatrix},$$

hence the equation $\nabla f(a,b,c) = 0$ has the solution $a = b = c = 0$. The Hessian matrix is

$$\nabla^2 f(a,b,c) = 2 \begin{bmatrix} b & a & c \\ a & c & b \\ c & b & a \end{bmatrix},$$

which is such that $\nabla^2 f(0,0,0) = 0$. [4 marks]

ii) Let, for example, $a = 0$, $b = \varepsilon$, and $c = \varepsilon$. Then $f(0,\varepsilon,\varepsilon) = \varepsilon^3$. Hence, $P_1 = (0,\varepsilon,\varepsilon)$ and $P_2 = (0,\varepsilon,-\varepsilon)$, with $\varepsilon > 0$, are such that

$$f(P_1) > f(P) = 0 > f(P_2),$$

which implies that P is a saddle point. [4 marks]

c) i) Note that

$$\nabla f(x) = 9 - \frac{4}{x-7},$$

and that

$$\nabla^2 f(x) = \frac{4}{(x-7)^2} > 0,$$

which means that the function is convex (in its domain of definition). In addition

$$\lim_{x \to 7^+} f(x) = \lim_{x \to \infty} f(x) = +\infty,$$

that is the function its radially unbounded in its domain of definition. The only stationary point of the function is obtained solving the equation $\nabla f(x) = 0$, which yields $x_\star = 7\,\frac{4}{9}$. This is therefore the global minimizer of f. [4 marks]

ii) Newton's iteration is

$$x_{k+1} = x_k - \frac{(9x_k - 67)(x_k - 7)}{4}.$$

Starting from $x_0 = 7.5$ one has

$$x_1 = 7.4375, \quad x_2 = 7.4443359375, \quad x_3 = 7.444444417 \approx 7.\dot{4} = 7\,\frac{4}{9},$$

which converges (with quadratic speed of convergence) to $x_\star$, whereas starting from $x_0 = 8$ one has

$$x_1 = 6.75, \quad x_2 = 6.359375, \quad x_3 = 4.795349121093750,$$

Answers

which arre out of the domain of definition of the function f and diverges. This is consistent with the fact that Newton's iteration yields only local convergence results (even for convex functions). Note also that Newton's iteration is well defined even outside the domain of definition of f.

[4 marks]

Answers